RAHUL KP

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SUMMARY

Software Engineer with an M.Sc. in Computer Science and hands-on experience building **backend APIs, full-stack applications, data-intensive systems, and AI/ML applications**. Strong in Python, FastAPI, PostgreSQL, Redis, React/Next.js, Docker, Git, and CI/CD. Experienced in designing modular application architectures, integrating services through REST APIs, implementing authentication and role-based access, and building deployable systems. Strong CS fundamentals with a focus on **reliable, maintainable, and production-oriented software**.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL **Backend:** FastAPI, REST APIs, PostgreSQL, Redis, SQLAlchemy, Pydantic **Frontend:** React, Next.js, TypeScript, Tailwind CSS **Engineering:** Data Structures & Algorithms, OOP, System Design, API Integration, Testing, Debugging **DevOps:** Git, GitHub Actions, Docker, Docker Compose, Nginx, Kubernetes, Terraform **AI/Data:** NumPy, PyTorch, Scikit-learn, Machine Learning, Deep Learning, NLP, LLMs, RAG **Databases:** PostgreSQL, MongoDB

PROJECTS

RankScript — Full-Stack Learning & Ranking Platform

FastAPI · PostgreSQL · Redis · Next.js · TypeScript · Docker

- Built a full-stack platform with **Student, Mentor, and Admin** roles, authentication, RBAC, course management, quizzes, assignments, and analytics.
- Designed modular REST APIs using a **Route → Service → Model** architecture with FastAPI, SQLAlchemy, Pydantic, and PostgreSQL.
- Implemented competitive ranking and leaderboard logic using quiz performance, assignments, course completion, and learning activity.
- Integrated **Redis caching, Docker containerization, Nginx, and CI/CD** for deployment and operational reliability.

NCF Recommender System — End-to-End Recommendation Platform

Python · NumPy · PyTorch · FastAPI · PostgreSQL · Docker

- Designed and implemented a recommendation system from **first principles**, including a NumPy implementation and PyTorch benchmark.
- Built an end-to-end architecture for data processing, model training, recommendation generation, API serving, and deployment.
- Implemented hybrid recommendation strategies to address **cold-start and sparse-interaction** scenarios.
- Achieved **0.6293 Hit@10 and 0.3541 NDCG@10** on evaluation experiments.

ANN Foundation — Neural Network & Automatic Differentiation

Python · Automatic Differentiation · Neural Networks · Testing

- Implemented a neural-network framework from scratch with **reverse-mode automatic differentiation and dynamic computation graphs**.
- Built composable Neuron → Layer → MLP abstractions with gradient accumulation and reverse topological backpropagation.
- Developed gradient verification tests using numerical differentiation and automated testing through GitHub Actions.

Phishing Detection — Machine Learning Security System

Python · Scikit-learn · Machine Learning

- Developed a machine-learning pipeline for detecting phishing URLs using feature engineering, model training, and evaluation.
- Applied classification techniques to identify malicious URL patterns and evaluated model performance using standard ML metrics.

EDUCATION

M.Sc. Computer Science — University of Calicut | 2024–2026 **B.Sc. Computer Science** — University of Calicut | 2020–2023

RESEARCH

Neural Collaborative Filtering Recommender System Research/preprint work focused on neural recommendation architectures, first-principles implementation, NumPy/PyTorch benchmarking, hybrid recommendation strategies, and reproducible evaluation.